EXPLORATORY DATA ANALYSIS REPORT

Workforce Performance & Absenteeism Analysis

Mega Supermart: workforce composition, tenure and age structure, absenteeism patterns and operational risk

Prepared by **Keren Oyem**, Business Intelligence Consultant | Submitted 09 November 2026

Executive Summary

Mega Supermart asked for an exploratory review of its workforce to support decisions on staffing, scheduling and human-capital risk. We analysed an anonymised dataset of **1,200 employees across 11 stores in four western provinces**, covering demographics, job attributes, length of service, scheduled hours and absent hours for 2025. The headline is that **absenteeism is not a company-wide problem; it is concentrated in specific departments, a small number of stores, and the first year of employment**.

1,200	38 yrs	3.1 yrs	2.5%	27 FTE
employees, 11 stores	median age	median service	absence rate (of scheduled hrs)	absent all year (55,984 hours)

Key findings

- **Workforce shape.** Sales is 37% of staff, Inventory 19% and Customer Service 16%. Four stores (Calgary, Edmonton, Vancouver, Surrey) hold 64% of employees, while Brandon, Regina and Saskatoon together hold only 8%. Gender is balanced (49% men, 47% women) and 72% work full-time.
- **Imbalances sit in the small stores.** Brandon and Saskatoon each have just one Customer Service employee, Regina has no HR staff, and Saskatoon has one supervisor for every 7.3 employees (about 3.4 elsewhere).
- **Retirement and turnover risk are in different places.** 8% of staff are 55+ (99 people), concentrated in HR (21%) and Operations (15%). About 17% have under one year of service (after removing suspect records), concentrated in Sales and Customer Service.
- **Absence is skewed and concentrated.** The median employee misses 32 hours but the mean is 47 because of a small group with very long absences. The top 10% of employees account for 36% of all absent hours. Customer Service (63 h) and Sales (57 h) are more than double Operations (26 h), and together generate about 67% of absent hours.
- **Regina is the standout store.** It averages 91 hours per employee, about 41 hours above what its department mix would predict, and 20% of its staff are extreme cases. Brandon (+13 h) and Winnipeg (+11 h) also run above expectation.
- **Age and gender do not explain absence; early tenure does.** Age and length of service are only weakly correlated with absence, and gender and full-time versus part-time show no difference. Employees under 25 (67 h) and in their first year (genuine new hires 59 h versus 42 h for established staff) are the groups at risk, especially in Sales.
- **Data issues need client attention.** Two employee IDs are shared by different people, 55 employees have placeholder hire dates (30 to 31 Dec 2025), and the job title "Store Manager" appears 58 times for 11 stores.

Priority actions. (1) Regina turnaround plan and HR cover. (2) First-year onboarding in Sales and Customer Service. (3) Return-to-work management for the roughly 50 long-term absence cases. (4) Rebalance coverage in small Prairie stores. (5) Succession planning in HR, Operations and Inventory. Each opportunity is worth about 6% to 9% of absent hours, so the value lies in **targeting**, not broad company-wide programmes.

1. Scope, Data and Approach

Data. One Excel workbook with an employee sheet (1,200 rows, 17 fields) and a data dictionary. Fields cover store, city, province, business unit, division, department, job title, gender, birth and hire dates, age, length of service, employment type, 2025 scheduled hours, absent hours and absence rate. Age and service are measured at 31 December 2025.

Approach. We followed an eight-step workflow: data understanding, data-quality review and cleaning, descriptive analytics, workforce composition, tenure and age, absenteeism, risk identification, and recommendations. All work was done in Python (pandas, seaborn, scipy) in a single reproducible Jupyter notebook that accompanies this report.

Definitions and assumptions. The brief does not define these terms, so we set them explicitly:

- **Retirement watch** is age 55+ and **high retirement risk** is 60+. **Turnover watch** is under one year of service.
- **Extreme absence** is any employee above the IQR upper fence (Q3 + 1.5 x IQR), which is 142 hours.
- **Absence rate** is absent hours divided by scheduled hours. **FTE** equals 2,080 hours.
- Findings are reported for groups only. Stores with fewer than 50 employees (Brandon, Regina, Saskatoon) are flagged because their averages are less stable.

2. Data Quality: Issues Identified and Clean-Up Steps

The data is generally clean: the age, tenure and absence-rate fields recompute correctly from the raw dates and hours, there are no negative values, no hires under age 18 and no fully duplicated rows. We found eight issues. **No rows were deleted**: all 1,200 records remain, with each problem corrected, flagged or deliberately left alone.

#	Issue found	Rows	Action taken
1	Department typed inconsistently ("sales", "Customer service")	2	Standardised labels
2	Department contradicts job title (a Customer Service Supervisor tagged "sales" and the reverse)	2	Department re-assigned from job title; flagged for client confirmation
3	Two EmployeeIDs each shared by two different people (different store, title, birth date)	4	Kept all records; created a unique RecordID
4	Missing Gender	3	Labelled "Not recorded" (kept distinct from "Prefer not to say")
5	Missing absent hours and absence rate	2	Not imputed (would invent behaviour); excluded from absence analysis only
6	Extreme absent hours (above 142 h; maximum 720 h)	50	Kept and flagged; results shown with and without them
7	Managers or assistant managers aged under 25 (possible title coding issue)	13	Kept and flagged for client validation
8	Hire date of 30 to 31 Dec 2025 (zero tenure) despite full-year hours and absence recorded; December shows 65 hires versus about 17 in a normal month	55	Treated as placeholder dates; kept, flagged, and tenure findings re-tested without them

Other checks. Age and scheduled hours have no outliers. The 52 tenure outliers are simply long-serving employees (up to 27 years) and are legitimate. Data types were corrected (dates and categorical fields), and bands for age and tenure were added. Separately, there are 58 "Store Manager" titles for only 11 stores, so the title probably refers to department-level managers; this should be clarified because it affects manager-ratio calculations.

3. Key Insights and Supporting Visuals

3.1 Workforce composition

Sales is by far the largest department (445 employees, 37%), larger than Inventory (228) and Customer Service (191) combined. Operations has 139, Management 129 and HR 68. Store size is very uneven: Calgary has 225 employees and Saskatoon 22.

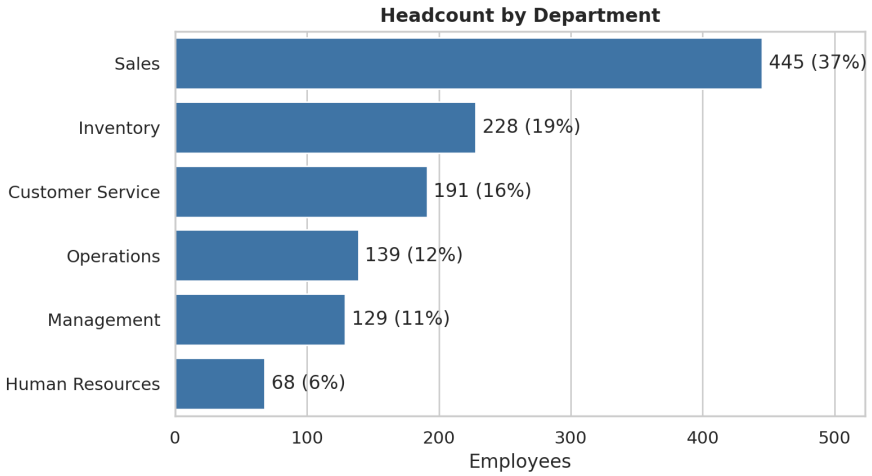


Figure 1. *Headcount by department.*

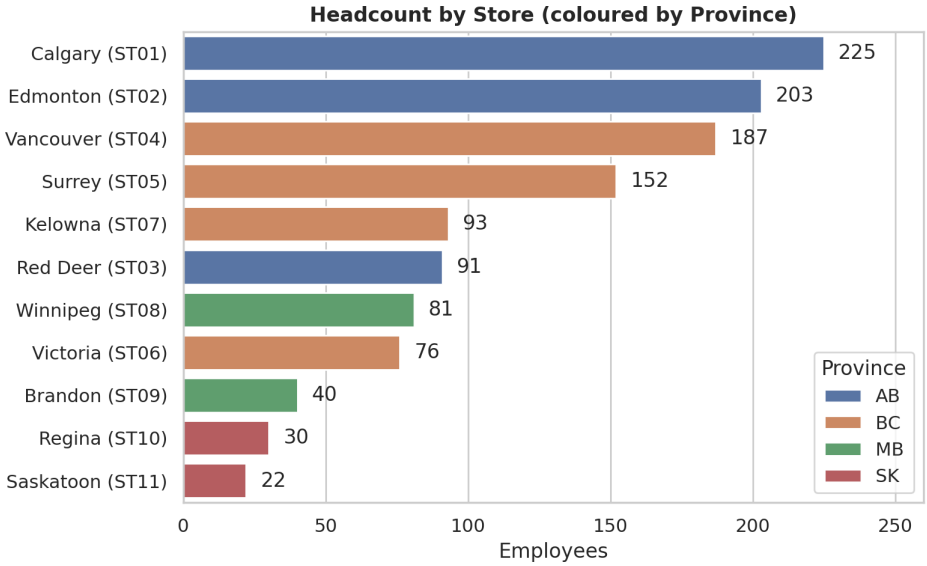


Figure 2. *Headcount by store, coloured by province. Four stores hold 64% of staff.*

The Prairie business unit has 692 employees (58%) and Pacific 508 (42%); the West division holds 86% of staff and Central 14%. Gender is evenly split overall and within every department (widest gap: HR, 53% women versus 43% men). Roughly 39% of employees hold a supervisory or management title (338 supervisors and 129 managers), which suggests either a supervision-heavy structure or generous title use.

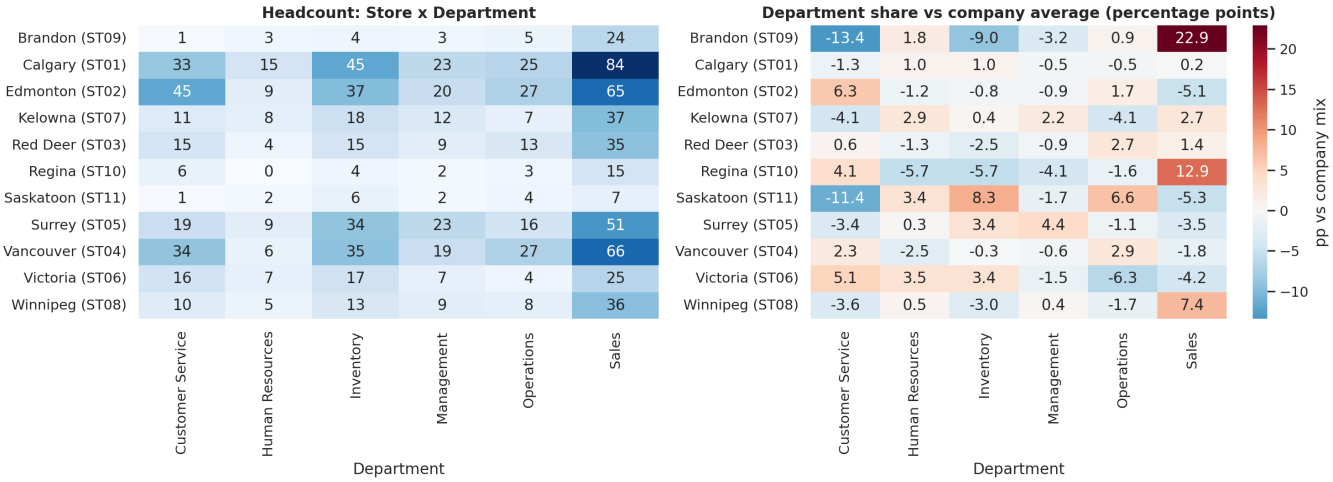


Figure 3. *Department headcount by store (left) and each department's share of the store versus the company average, in percentage points (right).*

Imbalances. Large stores mirror the company mix closely. The exceptions are small: Brandon is 60% Sales (23 points above the company mix) with a single Customer Service employee; Saskatoon also has one Customer Service employee and 7.3 staff per supervisor; Regina has no HR staff; Edmonton is relatively Customer Service-heavy.

3.2 Tenure and age structure

Age is bell-shaped around a median of 38 (range 18 to 66). Length of service is right-skewed: median 3.1 years, mean 4.5, maximum 27, with only 11% of staff at 10+ years. The company relies on a recent workforce with a small experienced core.

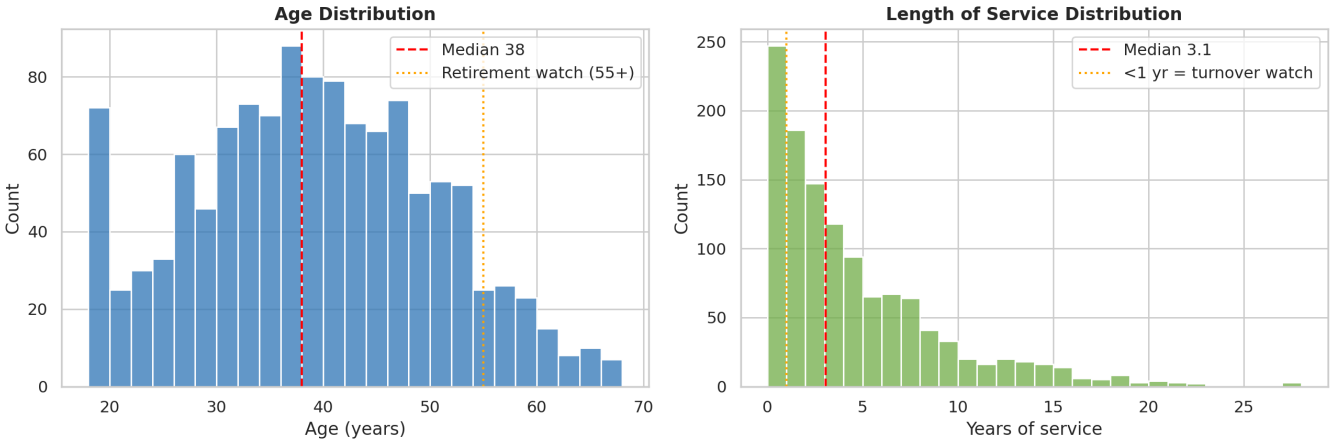


Figure 4. *Distribution of age (left) and length of service (right), with watch thresholds.*

Back-of-house departments are older and more tenured (Operations mean age 42.8, HR 42.5, Inventory 41.7; about 6 years of service). Front-line departments are younger and newer (Sales and Customer Service about 36 years old, 3.3 to 3.5 years of service). Age and tenure are only weakly related (r about 0.20), so we treat them as separate risk lenses.

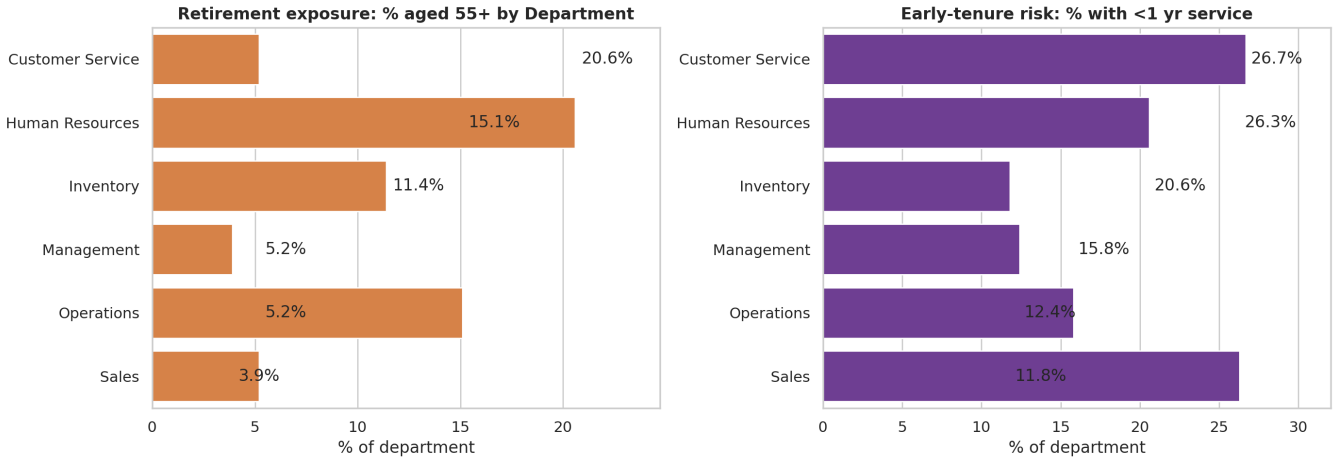


Figure 5. *Retirement exposure (% aged 55+) and early-tenure exposure (% under one year of service) by department. These are opposite pictures.*

Retirement. 99 employees (8%) are 55+ and 40 are 60+. In percentage terms HR (21%), Operations (15%) and Inventory (11%) are most exposed; Inventory has the most people (26). HR Business Partners (25% aged 55+) stand out among job titles. Only 5 store or assistant managers are 55+, so the management pipeline is relatively young.
Turnover. 247 employees (21%) have under one year of service, falling to about 17% after removing the 55 placeholder dates; it is highest in Customer Service and Sales (about 21% each).

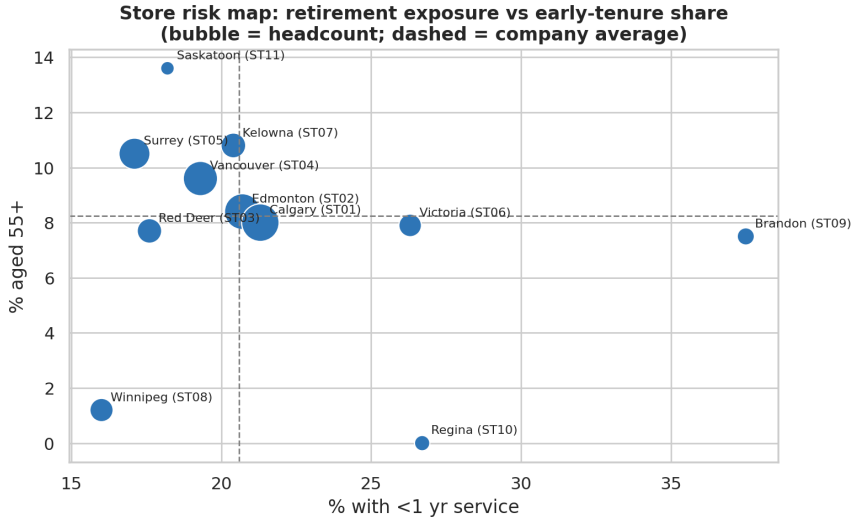


Figure 6. *Store risk map: retirement exposure versus early-tenure share (bubble size = headcount; dashed lines = company average).*

Brandon has 37.5% of staff under one year of service, well above the 21% company rate; Saskatoon, Surrey and Kelowna have the oldest workforces. Winnipeg and Regina have almost no retirement exposure. Brandon, Regina and Saskatoon are small, so these are signals to investigate rather than proven problems.

3.3 Absenteeism

Absence is strongly right-skewed. The median employee misses 32 hours (about four working days) but the mean is 47; 18% of employees recorded no absence at all, and the longest absence is 720 hours. The company-wide rate is 2.5% of scheduled hours, equal to about 27 full-time positions absent for the whole year. The top 10% of employees account for 36% of all absent hours, and the 50 extreme cases alone for about 21%.

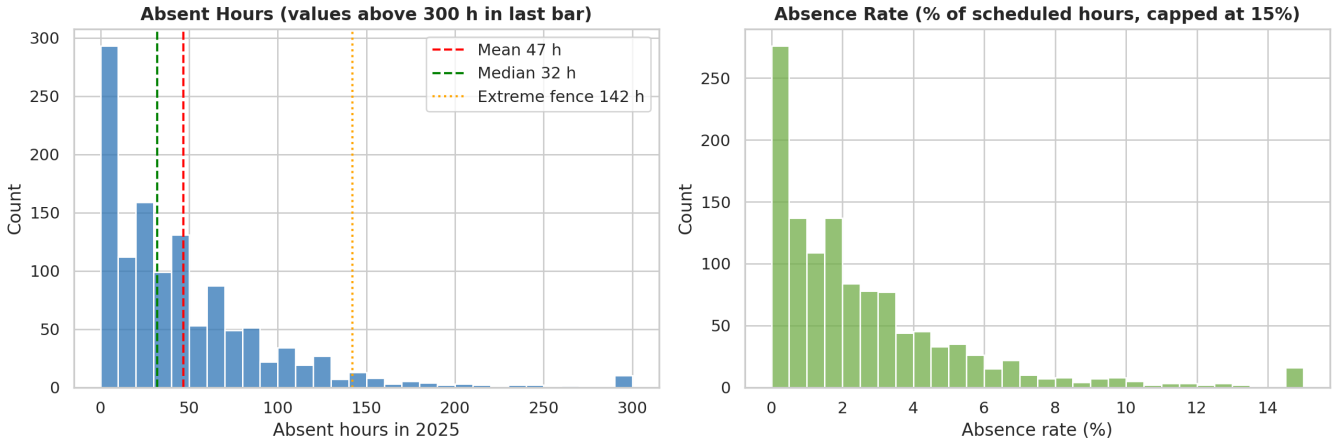


Figure 7. *Distribution of absent hours (left) and absence rate (right).*

Department and job title. Customer Service (mean 63 h, median 48 h) and Sales (57 h, 40 h) are highest; Operations (26 h), Management (30 h) and Inventory (34 h) are lowest. Sales and Customer Service make up 53% of staff but about 67% of absent hours. The highest-absence job titles are Customer Service Supervisor (67 h; 15% above the extreme fence), Customer Service Associate (66 h) and the Sales roles (56 to 59 h).

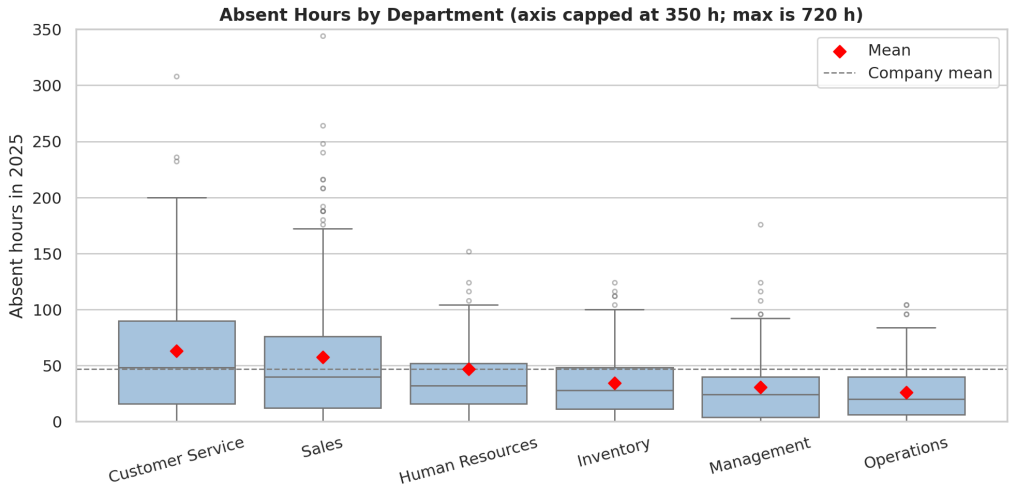


Figure 8. *Absent hours by department (red diamonds = mean; axis capped at 350 h).*

Stores and geography. Regina averages 91 hours per employee, nearly double the company mean, with Brandon (62 h) and Winnipeg (58 h) next. The large Alberta stores are lowest (about 41 to 42 h). By province, Saskatchewan (73 h) and Manitoba (59 h) sit well above Alberta (42 h) and British Columbia (46 h).

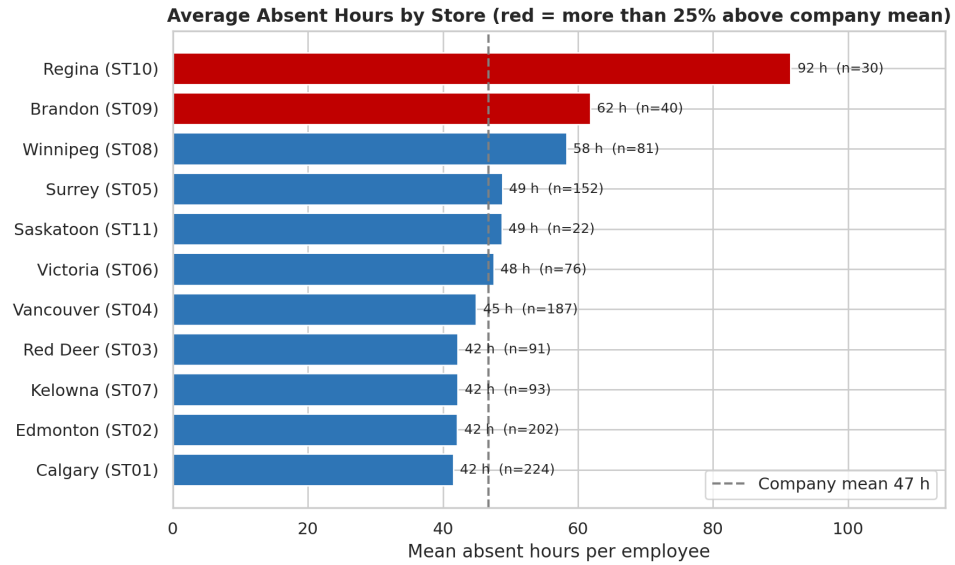


Figure 9. *Average absent hours by store (red = more than 25% above the company mean; n = employees).*

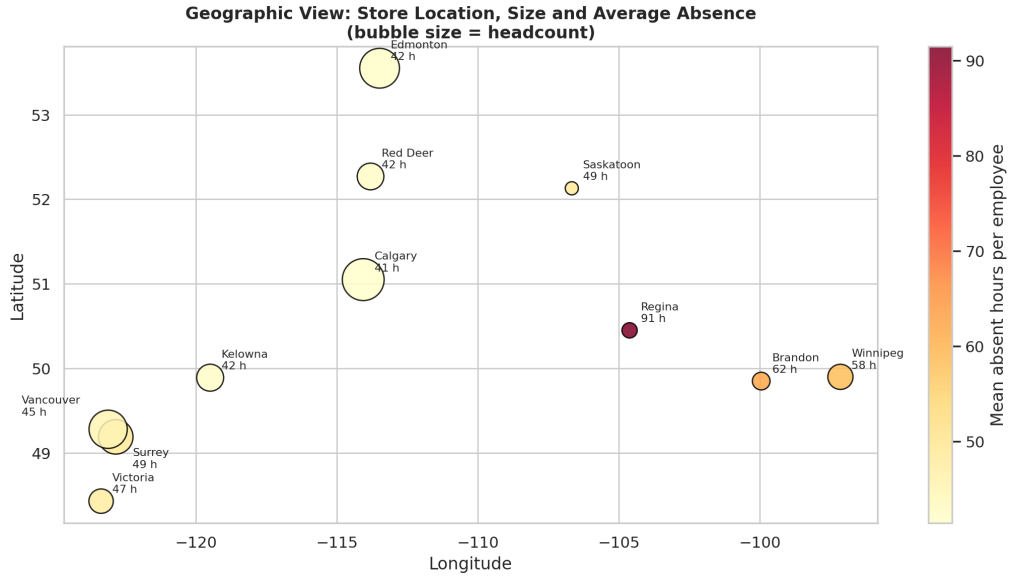


Figure 10. *Geographic view: store location, size and average absence (approximate city coordinates).*

Correlations. Age and absence are almost uncorrelated (r = -0.06 to -0.08), and length of service has a weak negative relationship (r = -0.12): longer-serving staff are slightly less absent. These are statistically detectable but small, so age and tenure alone do not explain absenteeism.

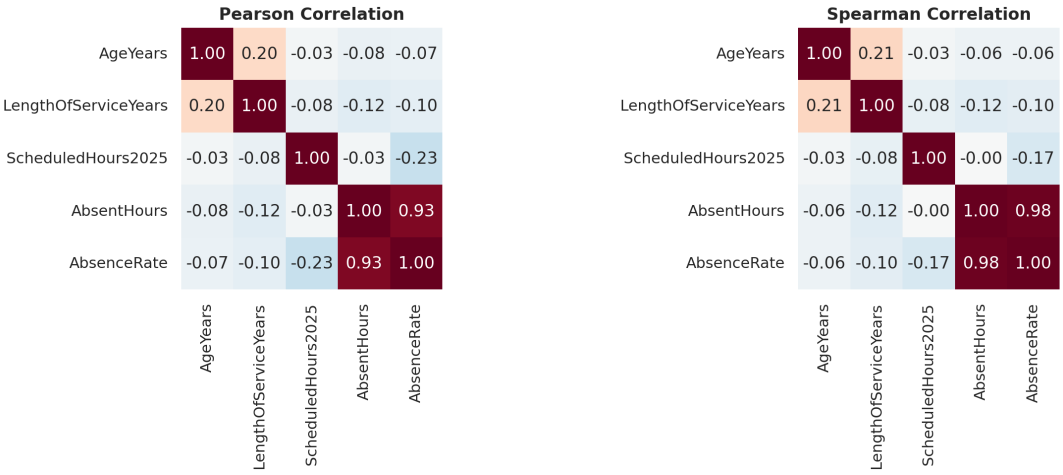


Figure 11. *Pearson (left) and Spearman (right) correlations between age, service, scheduled hours, absent hours and absence rate.*

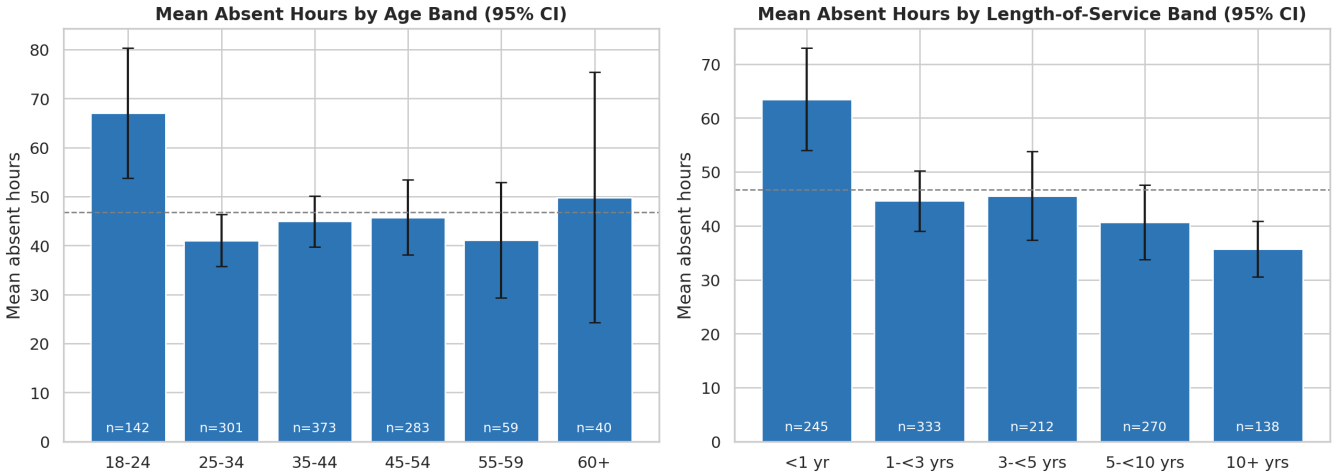


Figure 12. *Mean absent hours by age band (left) and length-of-service band (right), with 95% confidence intervals.*

Two segments stand out: **employees aged 18 to 24 (67 h)** and **first-year employees (63 h)**, against a company mean of 47 h. After the first year absence settles at 36 to 46 hours and falls to 36 h for staff with 10+ years. Employees aged 55+ are not more absent than average. The pattern points to an **onboarding and early-career engagement issue**, not an ageing issue.

Statistical tests. Because absence is skewed we used non-parametric tests (Kruskal-Wallis, Spearman). Department, job title and tenure band show the strongest differences ($p < 0.001$), followed by province, age band and store ($p < 0.05$). **Gender ($p = 0.68$) and employment type ($p = 0.86$) show no significant difference.** Effect sizes are small to moderate, so these factors shift averages but do not predict any individual's absence.

3.4 Is it the store, or the mix of departments?

Part of a store's absence is simply explained by its department mix, because Sales and Customer Service naturally miss more hours. We therefore compared each store with what its department mix would predict. **Regina remains about 41 hours per employee above expectation**, Brandon about +13 and Winnipeg about +11, so these are genuine store-level issues. Calgary, Edmonton, Red Deer and Kelowna run about 5 hours below expectation, which suggests better practice that could be shared.

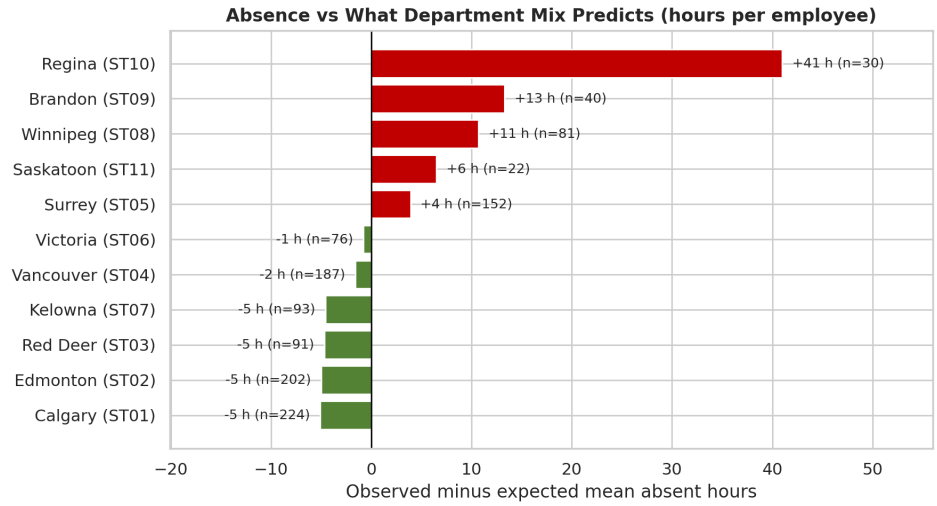


Figure 13. *Observed minus expected mean absent hours by store, after adjusting for department mix.*

The first-year effect also holds within departments. In Sales, genuine new hires average about 78 hours versus 49 for established staff, and under-25s average 74 versus 54. In Customer Service the gaps are small (64 versus 62 h), meaning absence there is high for everyone. Removing the 50 extreme cases lowers the company mean from 47 to 38 hours but **does not change the department ranking**, so the main conclusions are robust. The 55 suspect hire dates also do not drive the first-year finding: genuine new hires average 59 hours versus 42 for established staff.

3.5 Risk zones

Absence is concentrated in a handful of store-department teams, not spread evenly. Three teams (at least 8 employees each) run at 1.5 times the company mean or more: **Winnipeg Customer Service (151 h, 20% extreme cases)**, **Regina Sales (108 h, 27% extreme)** and **Brandon Sales (72 h)**.

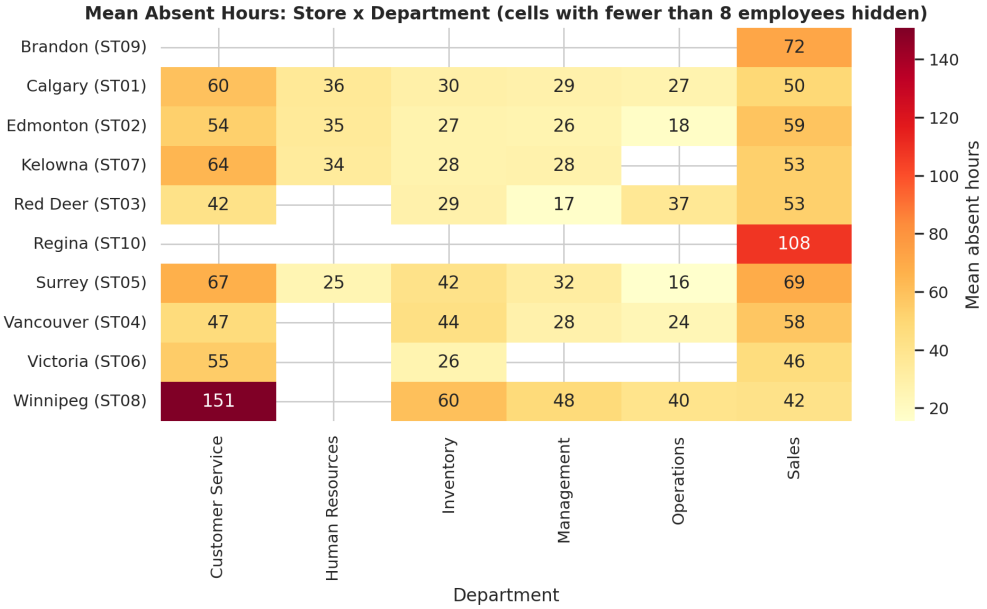


Figure 14. *Mean absent hours by store and department (teams with fewer than 8 employees hidden).*

Tenure extremes. Brandon Sales (42% under one year), Kelowna Customer Service (36%), Victoria Sales (36%) and Red Deer Sales (mean tenure 2.5 years) are low-tenure teams under strain. Edmonton HR (8.4 years), Kelowna Management (7.5) and Surrey Operations (7.5) are very stable, which is a strength but also a knowledge-concentration and retirement-planning risk.

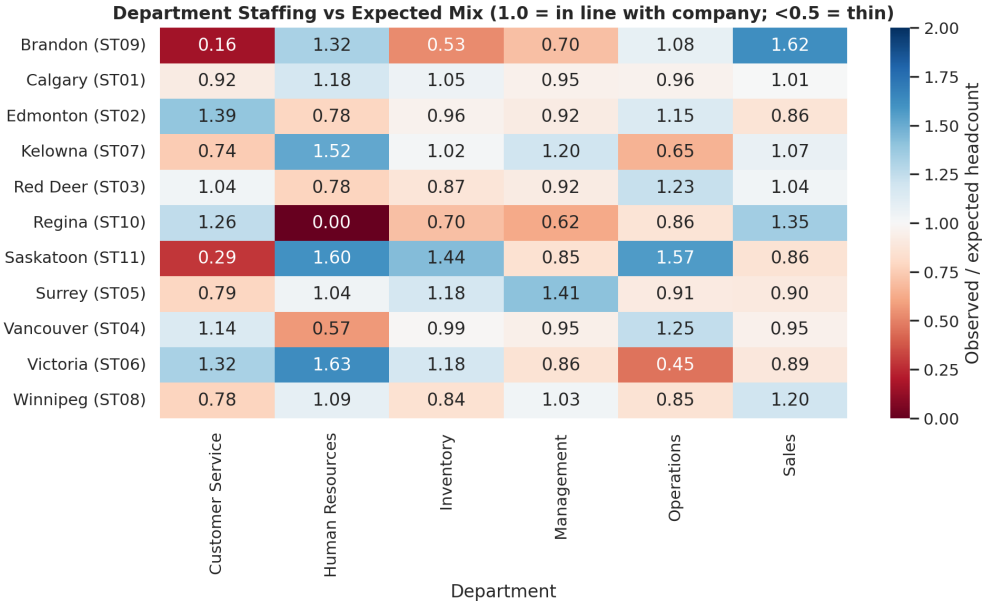


Figure 15. *Observed versus expected department headcount by store (1.0 = in line with the company mix; below 0.5 = thin).*

Possible understaffing. Against the company-wide mix, the clearest gaps are Brandon Customer Service (1 employee against about 6 expected), Saskatoon Customer Service (1 against 3.5), Victoria Operations (4 against 9) and Regina HR (none). This benchmark is an assumption, since a small store might legitimately combine roles, so these are questions for store managers rather than conclusions.

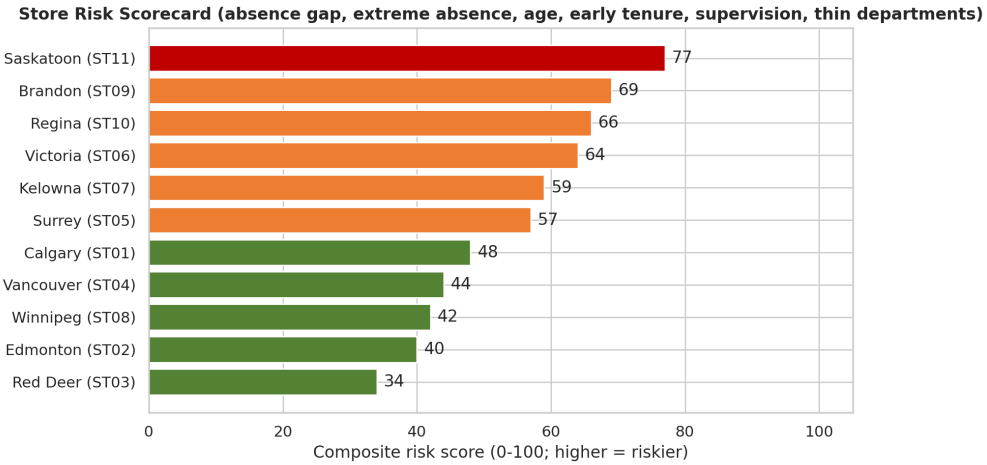


Figure 16. *Composite store risk score (absence gap, extreme absence, age, early tenure, supervision span, thin departments).*

Saskatoon (77), Brandon (69), Regina (66) and Victoria (64) rank highest; Edmonton (40) and Red Deer (34) rank lowest. The score uses equal weights and ranks, and the top three stores are small, so it should be used to decide where to look first rather than as a verdict. Regina has the worst absence problem but ranks third because it has no retirement exposure and dense supervision.

4. Answers to the Business Questions

Q1. Workforce composition: how is the workforce distributed, and are there imbalances?

The workforce is 1,200 people in 11 stores, led by Sales (37%), Inventory (19%) and Customer Service (16%). Alberta and BC hold 86% of staff; Manitoba and Saskatchewan hold 14%. The Prairie business unit has 58% of staff and the West division 86%. Gender is balanced (49% men, 47% women, 4% other or not stated). Yes, there are imbalances, and they sit in the small Prairie and Central stores: one Customer Service employee each in Brandon and Saskatoon, no HR in Regina, an unusually wide supervisor span in Saskatoon, and a Sales-heavy Brandon.

Q2. Tenure and age: what are the distributions and which segments are at risk?

Median age is 38 and median service 3.1 years (mean 4.5, right-skewed). Operations, HR and Inventory are older and more tenured; Sales and Customer Service are younger and newer. **Retirement risk:** 99 employees aged 55+ (40 aged 60+), concentrated in HR (21%), Operations (15%) and HR Business Partners (25%). **Turnover risk:** about 17% have under one year of service, concentrated in Sales and Customer Service and in Brandon (37.5%), Regina and Victoria (about 26%).

Q3. Absenteeism: what is the distribution, who is most affected, and what is it correlated with?

The median is 32 hours and the mean 47, with a long right tail (maximum 720). Customer Service, Sales, Regina, Brandon and Winnipeg, employees under 25 and first-year employees are highest. Absence correlates only weakly with age ($r = -0.06$) and service ($r = -0.12$). It differs significantly by department, job title and tenure band, and modestly by province, age band and store, but **not by gender or full-time versus part-time**.

Q4. Risk identification: which outliers, data issues and risk patterns exist?

Outliers: 50 employees above 142 absent hours (92% of them in Sales or Customer Service). **Data quality:** duplicate IDs, placeholder hire dates, label conflicts, an unclear Store Manager definition and a few missing values. **High-risk patterns:** absence clusters in Winnipeg Customer Service and Regina and Brandon Sales; low-tenure teams in Brandon, Kelowna, Victoria and Red Deer; possible understaffing in Brandon and Saskatoon Customer Service, Victoria Operations and Regina HR; and retirement exposure in HR and Operations.

Q5. Strategic opportunities: where can operations improve?

Three areas stand out: store-level recovery where absence exceeds the department-mix expectation (about 3,400 hours, 6% of absent hours); first-year onboarding in Sales and Customer Service (about 3,600 hours, 6.5%); and return-to-work management for extreme cases (about 4,900 hours, 8.7%). These overlap and should not be added together. The total absence burden is about 27 FTE, so no single action is transformational; the value comes from targeting and from reducing the disruption and cover cost of absence in specific stores.

5. Risks Identified

Risk	Where	Evidence	Level
Chronic high absence	Regina (all), Winnipeg Customer Service, Regina and Brandon Sales	Regina 91 h (+41 h vs expected, 20% extreme); Winnipeg CS 151 h	**High**
Early-tenure absence and churn	Sales and Customer Service; Brandon, Victoria, Kelowna	New hires 59 h vs 42 h established (78 vs 49 h in Sales); 17% under one year	**High**
Thin coverage and supervision span	Brandon and Saskatoon Customer Service; Regina HR; Saskatoon supervision; Victoria Operations	1 CS employee in each of two stores; no Regina HR; 7.3 staff per supervisor in Saskatoon	**High**
Long-term absence concentration	50 employees, mostly Sales and Customer Service	21% of all absent hours from 4% of staff	**Medium**

Risk	Where	Evidence	Level
Retirement and knowledge loss	HR, Operations, Inventory; HR Business Partners	21% / 15% / 11% aged 55+; 5 managers aged 55+	Medium
Data integrity	Employee ID, hire dates, titles	2 duplicate IDs; 55 placeholder hire dates; 58 Store Manager titles	Medium
Small-store fragility	Brandon, Regina, Saskatoon (22 to 40 staff)	Little depth to cover absence; volatile averages	Watch

6. Recommended Actions

Actions are ordered by expected impact. Owners are HR, store managers (SM) and operations leadership (Ops).

#	Action	Owner	When	KPI
1	**Regina turnaround plan:** management review, absence-reason analysis, local support, and a shared regional HR partner to cover the missing HR presence	Ops + HR	0 to 3 months	Regina mean absent hours within 15% of company mean in 12 months
2	**First-year onboarding and support** in Sales and Customer Service: buddy system, 30/60/90-day check-ins, scheduling that suits younger workers	HR + SM	0 to 6 months	First-year absence; 12-month retention
3	**Targeted attendance programme** for Customer Service and Sales floors: shift flexibility, workload review, wellness support	SM	3 to 6 months	Department mean absent hours
4	**Return-to-work and long-term leave management** for the roughly 50 extreme cases (supportive, not disciplinary)	HR	0 to 3 months	Cases above 142 h; days to return
5	**Rebalance coverage in small Prairie stores:** Customer Service in Brandon and Saskatoon, Operations in Victoria, supervisor span in Saskatoon	Ops + SM	3 to 9 months	Staff per supervisor; department coverage ratio
6	**Succession and knowledge transfer** for HR, Operations, Inventory and HR Business Partners; name successors for critical roles	HR	6 to 12 months	Critical roles with named successor
7	**Share best practice** from Calgary, Edmonton, Red Deer and Kelowna, which run below expected absence	Ops	3 to 6 months	Cross-store practice reviews held
8	**Review the supervision structure:** about 39% of staff hold supervisor or manager titles; confirm intent and clarify the Store Manager definition	Ops + HR	6 to 12 months	Span of control by store
9	**Fix the data:** unique employee IDs, consistent department labels, true hire dates, absence-reason codes	HR / HRIS	0 to 3 months	Data-quality checks passed

Insights for each audience

- **HR:** focus on first-year experience and extreme-absence cases; start succession planning in HR, Operations and Inventory; repair the data issues listed above.
- **Store managers:** in Sales and Customer Service, watch new hires and under-25s closely; Regina, Brandon and Winnipeg should review team-level absence drivers.
- **Operations leadership:** prioritise Regina, rebalance small-store coverage, and replicate the practices of the lowest-absence Alberta stores.

7. Limitations and Next Steps

- **One year, current employees only.** The data has no leavers, so turnover is inferred from short tenure rather than measured, and absence trends over time cannot be shown.
- **No absence reasons.** We cannot separate illness, injury, leave or unauthorised absence; reason codes would sharpen the response considerably.
- **Small groups.** Brandon, Regina and Saskatoon, and many store-department teams, have too few employees for firm conclusions.
- **Association, not causation.** Group differences describe patterns; they do not prove why absence is higher.
- **Assumptions.** Risk thresholds (55+, under one year, 142 h) and the expected-staffing benchmark are our assumptions and should be validated with the client.
- **Next steps.** Confirm true hire dates and the Store Manager definition, add absence reasons and multi-year history, then build a monitoring dashboard for the KPIs above.

Accompanying files. Keren_Oyem_Workforce_Absenteeism_EDA.ipynb (full analysis notebook), Keren_Oyem_Analysis_Tables.xlsx (supporting tables) and Keren_Oyem_Charts.zip (all 22 charts).